Term 4 answer key

Formulas from memory (the pyramid and cone formulas are printed). Lengths, areas and volumes are rounded only where the question says so (1 decimal place, the nearest degree or the nearest cent).

WEEK	SESSION	Q1 A Triangles & similarity	Q2 B Area & volume	Q3 C Percentages & money	Q4 D Indices, formulas & rates	Q5 E Graphs, data & chance
1	1	15/17	42.4 cm	$450	17	1/2
	2	6 cm	513.1 cm^3	50% profit	8.2×10^2	−1
	3	7.0 m	153.9 m^2	$1317.31	1/20 or 0.05	11.8
2	1	36°	288 cm^3	$60	4 km/h	0.6
	2	11.7 cm	19.7 L	$570	$57 and $38	0.28 or 7/25
	3	16 cm	157.1 cm^3	$72.24	2^7	7
3	1	12.3 m	37.7 m^2	$162	157.5 km	−3
	2	54°	122 m^2	$6345	4.5×10^4	7/10
	3	3 cm	100 cm^3	$975	$1/3^4$ = 1/81	2/9
4	1	5.9 cm	127.2 m^2	300	0.8 kg and 1.6 kg	140
	2	20 cm	336 cm^3	$2265.38	y^5	gradient 3, y-intercept −2
	3	31.1 m	22.0 cm	$360	3.5	7
5	1	13 cm	39.3 cm^2	20% loss	2.85×10^7	0.875
	2	7/25	19.6 cm^2	$38.99	1/8	0.42 or 21/50
	3	4 m	8 cm	$1210.95	$25x^8$	gradient 0.5, y-intercept −3
6	1	54.0 m	189 L	$6706.90	4.5	2.9
	2	6.5 m	296.9 cm^2	3936	90 km/h	1/2
	3	15/8	628.3 cm^3	$51	$93.75 and $156.25	2
7	1	10 cm	168 cm^2	$1016.50	4.5×10^{-4}	150
	2	12.8 cm	56.5 m^3	25% loss	$6x^7$	4
	3	31°	30 L	$980	29	3
8	1	26 m	879.6 cm^2	$5170	15 km/h	30
	2	7.0 cm	160 m^2	$1200	$1/5^2$ = 1/25	2/11
	3	6 cm	400 cm^3	$1500	$37.50 and $62.50	0.64
9	1	19.8 cm	58.9 cm^2	$5162.48	35	3/4
	2	8 cm	47.1 cm^3	90	20 km/h	4
	3	28°	384.8 cm^3	$1156.25	3/2	5/6
10	1	8/17	15.7 cm	$360	$45 and $75	3/4
	2	9.2 m	110 cm^3	20% profit	$3p^5$	0.22 or 11/50
	3	15.4 m	176.7 cm^2	$1000	5000	13

NSW MATHEMATICS K–10 (2022) OUTCOMES PRACTISED · STAGE 5, WITH STAGE 4 REVISION · MAO-WM-01 THROUGHOUT

A MA4-PYT-C-01 Pythagoras' theorem; MA5-TRG-C-01 trigonometric ratios (sides and angles); MA5-TRG-C-02 angles of elevation and depression (Terms 3–4); MA5-GEO-C-01 similar figures

B MA4-ARE-C-01 area of circles and sectors*; MA4-LEN-C-01 arc length*; MA5-ARE-C-01 surface area of prisms and cylinders; MA5-VOL-C-01 volume of prisms and cylinders, capacity; MA5-VOL-P-01 (Path, Term 4 only) volume of pyramids and cones

C MA5-FIN-C-01 simple interest, earning money (wages, overtime, salary, commission, leave loading); MA4-FRC-C-01 percentage increase and decrease, original amount, percentage profit and loss*

D MA5-IND-C-01 index laws with coefficients, zero and negative indices (numerical bases); MA5-MAG-C-01 scientific notation; MA4-ALG-C-01 / MA5-EQU-C-01 substituting into formulas and solving*; MA4-RAT-C-01 ratios and rates (speed)

E MA5-LIN-C-01 gradient and $y = mx + c$; MA4-DAT-C-02 mean, including frequency tables*; MA4-PRO-C-01 / MA5-PRO-C-01 probability, complementary events, relative and expected frequency*

* the code is right for the topic, but where this content sits within it could not be checked against the NESA website.